# Melanotan II (MT-II)

**100–250 mcg SC daily; 250–500 mcg SC daily or 1–2 times weekly**

**Melanotan II (MTII) has demonstrated skin-darkening and erectile effects in small human studies, with appetite reduction also reported during experimental administration. [1,2,16] It is a synthetic cyclic heptapeptide analogue of α-melanocyte-stimulating hormone (α-MSH). [1] It activates MC1R, MC3R, MC4R, and MC5R signaling, with MC1R supporting pigmentation and central melanocortin pathways contributing to feeding and erectile responses. [10,11] Nausea and other systemic effects have occurred in human testing. [1,2,16] Changing pigmented lesions, melanoma, and serious acute adverse events have been reported, with causal certainty differing by outcome. [5–8,17–19]**

☠️ Possibly unsafe; serious adverse events reported. [5–8,14,17–19]

⭐ ⭐ Likely effective for skin darkening (small human pilot study). [1]

Caution with a history of melanoma or atypical moles; avoid unsupervised use with active cancer. This is precautionary advice, not proof that MTII promotes every cancer. [5,6,9,14]

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

Nausea, fatigue, and erectile effects were observed during short human experiments. [1,2,16] Serious events after nonmedical exposure include priapism and systemic toxicity with muscle and kidney injury. [7,8] Melanoma reports constitute a dermatologic safety signal, not proof of carcinogenicity or a numerical risk estimate. [5,6]

Skin darkening was measured in a very small pilot study, while erections were documented in a small placebo-controlled trial. [1,2]

## Dose and Patterns

The human pigmentation pilot began at 0.01 mg/kg SC and escalated to 0.025–0.03 mg/kg, alternating MTII and saline over two weeks. [1] The organic and psychogenic erectile-dysfunction crossover studies used 0.025 mg/kg SC. [2,16] These historical exposures differ from the proposed loading and maintenance patterns above.

The following speculative experimental dose patterns are discussed in peptide communities. These patterns describe proposed nonmedical use.

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- **100–250 mcg SC daily for several days — initial pattern to test tolerability,**
- **250–500 mcg SC daily — loading pattern to achieve cosmetic effects,**
- **250–500 mcg SC once or twice weekly — maintenance pattern to maintain cosmetic effects.**

## Benefits

- **Likely visible skin darkening** — (small human pilot) Two of three men developed increased pigmentation after five MTII injections over two weeks; pigmentation was assessed one week after completion. The sample limits estimates of consistency, magnitude, and speed. [1]
- **Plausible partial photoprotection** — (mechanistic extrapolation) Increased pigmentation and melanocortin-related cellular repair provide a biologically plausible protective hypothesis. The cellular repair evidence involves α-MSH, and the human UV-response evidence below involves afamelanotide. [12,13]
- **Erections in men with erectile dysfunction** — (small controlled human trials) Ten men with organic risk factors participated in a crossover study; erections occurred after 12 of 19 MTII injections versus 1 of 21 placebo injections, and sexual desire increased. [2] In a separate crossover study of ten men with psychogenic erectile dysfunction, eight developed clinically apparent erections after MTII. [16] These findings concern the studied male populations.

## Investigated and Reported Uses

- **Skin darkening and erectile responses** — investigated in small human studies. Cosmetic tanning is a reported nonmedical use of an unapproved product; “off-label” should not imply that MTII itself is an approved medicine. [1,2,4]

## Cautions

- **Personal or family history of melanoma, or atypical moles** — *(precaution based on dermatologic concern)* changing lesions warrant assessment. [5,6]
- **Uncontrolled cardiovascular disease** — *(clinical precaution)* systemic toxicity with tachycardia and elevated blood pressure has been reported, including after a large dose. [7]
- **Pregnancy and breastfeeding** — *(safety not established)* the cited human studies do not support use in these populations. [1,2]

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## Side effects

- **Nausea, yawning, stretching, fatigue, and sleepiness** — (human studies) These effects occurred during short experimental exposures. **Severe nausea** followed 4 of 19 MTII injections in the organic erectile-dysfunction study. [1,2,16]
- **Spontaneous or prolonged erections** — (human trials and case reports) Erections may be unwanted. [1,2] A reported ischemic priapism episode required surgery after conservative measures failed, with persistent erectile impairment at 15 weeks. [8]
- **Appetite and cardiovascular effects** — Decreased appetite was reported during a controlled human experiment. [16] Tachycardia and hypertension occurred in a systemic toxicity case; these observations do not define the incidence at other exposures. [7]
- **Darkening or appearance of atypical moles** — *(case reports)* new or changing pigmented lesions have occurred after use. Melanoma has also been reported in association. [5,6]
- **Serious systemic events** — Rhabdomyolysis and kidney dysfunction were documented after a 6 mg injection, with MTII confirmed in the supplied product. [7] Posterior reversible encephalopathy syndrome (PRES) has been reported. [14,17] A renal-infarction case followed intermittent use, with the last reported injection three weeks before admission; the temporal association and product identification do not establish causation. [19]

## Clinical and Regulatory Status

MTII has early human research but no established cosmetic treatment regimen. FDA identifies potential immunogenicity and serious reported adverse events in its compounding safety assessment. [4,14]

## Biological mechanism

MTII is a cyclic α-MSH analogue. [1,10] It is a nonselective agonist at MC1R, MC3R, MC4R, and MC5R, with receptor experiments demonstrating cyclic-AMP signaling. [10] MC2R is the ACTH-responsive receptor and requires its accessory protein MRAP for ACTH binding and signaling. [20] MC1R supports pigmentation, while central melanocortin pathways provide a biological explanation for feeding and erectile effects. [1,2,11,16]

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## Pigmentation and UV Protection

MTII-induced pigmentation was observed in a three-person human pilot. That study did not measure a protective SPF or establish reduced skin-cancer incidence. Pigmentation should therefore be distinguished from a quantified clinical protection level. [1]

A related compound provides separate human evidence. In nine participants, afamelanotide reduced the measured UV-erythema response, while its increase in minimal erythema dose (MED) was not statistically significant. These are findings for afamelanotide, not MTII. [13]

Plausible protection remains a mechanistic interpretation. In cultured human melanocytes, α-MSH supported antiapoptotic pathways and reduced UV-related DNA damage. [12] Ordinary sun protection, including sunscreen, protective clothing, and shade, remains appropriate regardless of increased pigmentation. [15]

## Melanotan II and Cancer

MTII use has been associated with atypical moles and melanoma. These case reports cannot establish a numerical risk estimate. [5,6] Avoiding nonmedical cosmetic use is a precaution supported by the overall adverse-event concerns. [3,4,7,8,14]

## Safety Evidence summary

**Human dermatologic observations**: Published reports include melanoma in situ and rapidly appearing atypical moles after melanotan exposure. [5,6] Baseline susceptibility, UV exposure, coincidental disease, and uncertain product composition complicate causal attribution. [3,5,6]

Mechanistic evidence is not uniformly cancer-promoting. α-MSH reduced UV-related DNA damage in cultured human melanocytes. [12] Topical MTII reduced implanted melanoma growth in a mouse model, with PTEN-related changes, reduced AKT/NF-κB/COX-2/$PGE_2$ signaling, and changes consistent with reduced proliferation and angiogenesis and increased apoptosis. [9]

**Renal, muscular, and cardiovascular endpoints:** Acute muscle injury, kidney dysfunction, tachycardia, and hypertension occurred in a toxicity case. [7] A separate renal-infarction report described persistent hypertension and mildly reduced renal clearance at two months. [19] These are serious safety signals; neither common-event incidence nor a favorable cumulative-exposure prediction can be estimated from these reports.

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**Neurological and sexual endpoints:** Somnolence and fatigue occurred during short human testing. [1] A PRES report provides a serious neurological signal, while ischemic priapism can cause lasting erectile impairment. [8,17]

**Endocrine and metabolic endpoints:** A man with type 1 diabetes developed marked hyperglycaemia, ketosis without acidosis, hypertension, and transient hypercortisolism after intermittent use of a product labeled Melanotan 2. Product exposure was not analytically confirmed, and the case does not establish the proposed receptor or assay-interference explanations. [18]

## Practical recommendations

- **Avoid unsupervised cosmetic MTII use**, particularly with active cancer, prior melanoma, many atypical moles, or other substantial skin-cancer risk. This is precautionary advice rather than a validated drug-label contraindication list. [4–6,14]
- **After use, obtain clinical assessment of new or changing lesions.** Tell the clinician about MTII exposure and UV exposure; the clinician can determine whether a full skin examination and ongoing surveillance are appropriate. Monitoring does not make continued use safe. [5,6]
- **Do not use MTII as a skin-cancer prevention strategy.** Maintain ordinary sun protection and avoid intentional excess UV exposure. [4,12,15]

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